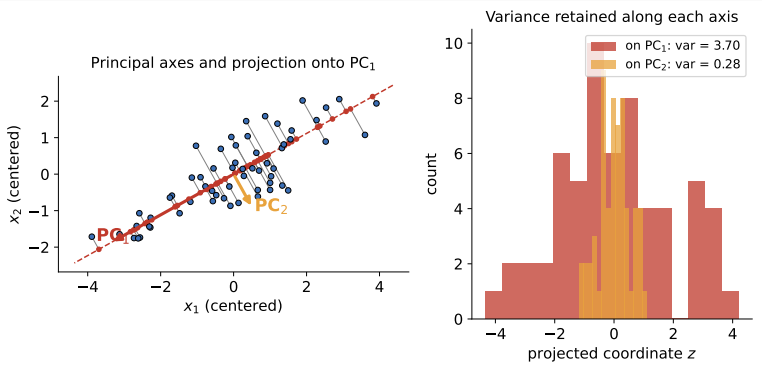
Principal Component Analysis

A short introduction

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Machine Learning Course

Principal Component Analysis (PCA): the idea



PCA finds new orthonormal axes, the **principal components**: PC_1 along the direction of largest variance, PC_2 along the largest remaining variance orthogonal to PC_1, ...

Keeping the first d PCs preserves as much spread as possible (here PC_1 keeps 93%). Equivalently, PC_1 minimizes the mean squared distance from points to their projections (grey segments).

PCA: the mathematics

1. Center the data ($\mathbf{X} \in \mathbb{R}^{m \times n}$, rows $\mathbf{x}^{(i)}$):

$$\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}^\top, \qquad \mathbf{C} = \tfrac{1}{m-1}\,\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}.$$

2. Maximize the projected variance:

$$\max_{\mathbf{w}} \ \mathbf{w}^\top \mathbf{C}\,\mathbf{w} \quad \text{s.t.} \quad \|\mathbf{w}\|_2 = 1.$$

Lagrangian $\mathcal{L} = \mathbf{w}^\top \mathbf{C}\mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{w} - 1)$, $\nabla_{\mathbf{w}} \mathcal{L} = 0$ gives

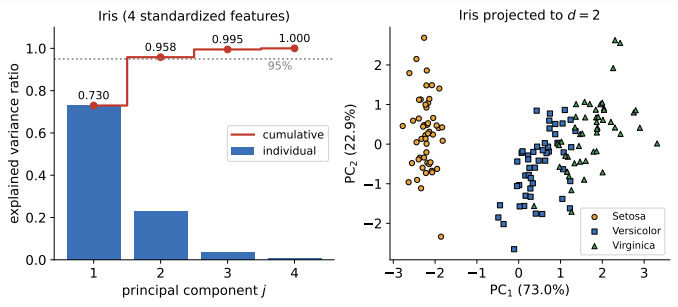
$$\boxed{\mathbf{C}\,\mathbf{w} = \lambda\,\mathbf{w}} \qquad \Rightarrow \qquad \mathbf{w}^\top \mathbf{C}\,\mathbf{w} = \lambda.$$

- The principal components are the eigenvectors $\mathbf{w}_1, \ldots, \mathbf{w}_n$ of $\mathbf{C}$, ordered by eigenvalue $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$.
- λ_j = variance of the data along $\mathbf{w}_j$.
- In practice: SVD $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$; the columns of $\mathbf{V}$ are the PCs and $\lambda_j = \sigma_j^2/(m-1)$.

3. Project onto the first d PCs:

$$\mathbf{Z} = \tilde{\mathbf{X}}\,\mathbf{W}_d, \qquad \mathbf{W}_d = [\mathbf{w}_1 \cdots \mathbf{w}_d].$$

Choosing the number of components



$$\mathrm{EVR}_j = \frac{\lambda_j}{\sum_{i=1}^{n} \lambda_i}, \qquad d^\star = \min\left\{ d : \sum_{j=1}^{d} \mathrm{EVR}_j \geq 0.95 \right\}$$

Iris: $d = 2$ keeps 95.8% of the variance, so 4 features become 2.

- **Standardize** the features first when their units differ.
- PCA is **linear** and **unsupervised**: it ignores the labels y.
- Reconstruction: $\hat{\mathbf{X}} = \mathbf{Z}\mathbf{W}_d^\top + \mathbf{1}\bar{\mathbf{x}}^\top$.